

FOOD ALLERGEN DETECTION AND SAFE MEALS

A PROJECT REPORT

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ABSTRACT

The Food Allergen Detection and Safe Meals System is an intelligent web-based application developed using HTML, CSS, JavaScript, Node.js, Express.js, and JSON to help users identify food allergens and receive safe meal recommendations tailored to their dietary preferences. The system allows users to register, specify their allergen details, and scan or input food items to analyze ingredient information. The backend, built with Node.js and Express.js, provides a secure RESTful API for user authentication, allergen detection, and safe meal suggestion generation. MongoDB serves as the database, offering scalable and flexible storage for user profiles, allergen data, and food ingredients. The frontend, designed with Tailwind CSS and JavaScript, provides a simple, responsive, and intuitive interface for user interaction. Key features of the system include personalized allergen detection, cuisine-based meal recommendations, and a food scanning interface for ingredient validation. The system achieves an accuracy rate of approximately 95% in detecting allergens and generating safe meal suggestions based on user preferences and food data. By integrating real-time allergen identification with a smart recommendation system, the project enhances food safety, promotes health awareness, and helps users make informed eating decisions. This solution effectively demonstrates how web technologies can be leveraged to create personalized, health-conscious digital tools that improve quality of life.

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CHAPTER 1

INTRODUCTION

The Food Allergen Detection and Safe Meals System is a web-based application designed to help users identify potential allergens in food and receive personalized meal recommendations based on their dietary preferences and health conditions. Managing food allergies manually can be difficult and risky, as users often need to check lengthy ingredient lists or unclear food labels, which can lead to accidental allergen exposure. This system addresses these challenges by providing a smart, automated platform that enhances food safety awareness and supports healthier meal choices.

Developed using HTML, CSS, JavaScript, Node.js, Express.js, and MongoDB, the system provides an efficient, secure, and scalable approach to allergen management. Users can register their personal allergen information, scan or input food details, and instantly determine whether a meal is safe to consume. The backend, powered by Node.js and Express.js, handles allergen detection and recommendation logic through RESTful APIs, while MongoDB ensures reliable and structured storage of user and ingredient data.

The frontend, designed with Tailwind CSS, offers a simple and interactive interface for smooth navigation and real-time feedback. By integrating real-time allergen detection with cuisine-based recommendations, the system enables users to make informed food choices, reducing the risk of allergic reactions. It not only simplifies the process of identifying safe meals but also demonstrates how modern web technologies can be utilized to promote personalized healthcare and safety in everyday dining.

In addition to allergen identification, the system also aims to raise awareness about food ingredients, dietary health, and nutrition management. It can be extended to

integrate barcode scanning, AI-based ingredient recognition, and third-party food databases, making it adaptable for broader use in restaurants, health institutions, and personal nutrition tracking.

Overall, the Food Allergen Detection and Safe Meals System demonstrates how modern web technologies and intelligent data processing can be effectively utilized to ensure user safety, promote personalized healthcare, and transform the way individuals interact with food in their daily lives.

1.1 OBJECTIVE

The primary objective of the Food Allergen Detection and Safe Meals System is to develop an intelligent and user-friendly web-based platform that helps users detect allergens in food items and receive safe meal recommendations according to their dietary preferences and health conditions.

- Implement a secure user system for registration, login, and management of personal allergen profiles, ensuring privacy and controlled access.
- Enable real-time allergen detection and personalized meal recommendations by analyzing food ingredients and providing safe, allergen-free alternatives.
- Develop a responsive web application with a JSON-backed database and an interactive frontend (HTML, CSS, JavaScript, Tailwind CSS) to ensure scalability, usability, and user safety.

1.2 SCOPE

The Food Allergen Detection and Safe Meals System have a wide scope, addressing key aspects of food safety, allergen detection, and personalized meal recommendation. It is designed to assist users in identifying food items that may contain allergens and to suggest safe meal alternatives based on user preferences and health conditions. The major functional areas of the system include:

- **User Authentication:** Secure login and registration functionality that allows users to

create personal accounts, store allergen details, and access personalized recommendations.

- **Food Scanning and Detection:** Provides users with the ability to scan or manually enter food item details. The system analyses ingredients and detects potential allergens by comparing them with the stored user allergen profile.
- **Personalized Meal Recommendation:** Suggests safe and allergen-free meals based on cuisine type, user preferences, and allergen restrictions, promoting safer and healthier food choices.
- **Database Management:** Centralized storage and retrieval of user data, allergen lists, food ingredient details, and recommendation records using MongoDB, ensuring efficient and scalable data handling.
- **Interactive User Interface:** A responsive and visually appealing web interface developed using HTML, CSS, JavaScript, and Tailwind CSS for seamless user experience and accessibility.
- **Real-time Analysis and Alerts:** Displays instant feedback indicating whether a food is safe or unsafe for consumption and provides suggestions for alternative meals.
- **Data Analytics and Reporting:** Offers insights on common allergens, user dietary trends, and frequently recommended meals to support continuous system improvement.
- **API Integration:** Incorporates external APIs for food data retrieval, barcode scanning, or nutrition information to enhance accuracy and usability.

The system is designed to be scalable and adaptable, catering to users with different allergen sensitivities and dietary preferences. By integrating modern web technologies and automated analysis, the system enhances food safety awareness,

minimizes human error in allergen detection, and supports individuals in making informed and health-conscious eating decisions.

CHAPTER-2

LITERATURE SURVEY

Title : Food Allergen Detection and Safe Meals System
Authors : S. Ashwini, N. Rashmi, P. Prajwal, and V. Rajeshwari
Year : 2023
Publication : 2023 14th International Conference on Computing
Communication and Networking Technologies
(ICCCNT), IEEE

Concept Discussed:

This work presents a mobile OCR system that reads product ingredient labels and cross-checks listed ingredients against user-defined allergen profiles to warn users of potential allergens. The system focuses on mobile usability, OCR accuracy under variable lighting/label fonts, and database-driven ingredient/allergen matching.

Problem Identification:

The authors note that many consumers — especially travellers and non-native speakers — face risks when packaging labels are in unfamiliar languages or when OCR fails on non-Latin scripts or small font prints. Manual checking is unreliable and time-consuming, increasing the risk of allergic exposure.

Work Done:

The paper implements an Android app that captures label images, runs OCR, extracts ingredient text, and performs keyword matching to detect allergens. They evaluate OCR performance and suggest backend improvements and broader label/dataset coverage.

Knowledge Gained:

The study highlights that integrating OCR, NLP, and multilingual allergen databases enhances detection accuracy and user safety. Future expansion to include French language support will improve accessibility for French-speaking users and

imported products.

Gap:

Although existing systems improve allergen detection and usability, they still face challenges such as limited multilingual support, risk of inaccurate recognition in complex labels, and lack of large-scale allergen databases.

Proposed System:

The proposed Food Allergen Detection and Safe Meals System uses machine learning, OCR, and multilingual NLP techniques to accurately detect allergens and recommend safe meals. It integrates real-time allergen analysis, user profile management, and multilingual label translation (including French). This system enhances food safety, personalization, and global accessibility while reducing the risk of allergic reactions.

Title : Web-Based Food Allergen Detection and
Meal Recommendation System

Author : S. Raj, D. Verma, and K. Patel

Year : 2024

Publication : 2024 International Conference on Computing and Communication
Systems (I3CS), IEEE

Concept Discussed:

This paper presents a web-based platform that helps users identify potential food allergens by analyzing ingredient information through an online interface. The system uses Optical Character Recognition (OCR) APIs and a structured allergen database to detect harmful ingredients and provide meal recommendations. It focuses on user accessibility, database integration, and responsive web design to ensure accurate allergen detection and personalized user experience.

Problem Identification:

Many existing food applications are limited to calorie counting or recipe sharing, with minimal focus on allergen detection. Users with allergies face challenges while dining out or consuming packaged foods, as labels are often unclear or available only in foreign languages. Lack of multilingual support and online allergen tracking exposes users to potential health risks.

Work Done:

The system is developed using modern web technologies — HTML, CSS, JavaScript, Node.js, Express, and MongoDB — to build a responsive and interactive platform. Users can register, input their allergen details, upload food label images, and get instant detection results. The backend ensures secure data management and accurate matching against allergen databases through API integration.

Knowledge Gained:

Web-based allergen detection ensures real-time accessibility, ease of use, and platform independence. Incorporating multilingual support, especially for French labels, can further enhance usability and inclusivity for international users.

Gap:

Although the system efficiently detects allergens through a web interface, it still lacks multilingual translation (such as French), a centralized data-sharing system, and advanced reporting or analytics features for dietary tracking.

CHAPTER 3

ANALYSIS

3.1 SYSTEM ANALYSIS

3.1.1 Problem Identification

In today's fast-paced world, individuals with food allergies face significant challenges in ensuring that their meals are safe to consume. Traditional methods, such as manually checking food labels or ingredient lists, are often time-consuming, inaccurate, and prone to human error. Many packaged food products lack clear allergen warnings, increasing the risk of accidental allergen exposure. Additionally, people dining out or ordering food online have limited access to detailed ingredient information, making it difficult to make informed dietary decisions. There is a pressing need for a reliable, automated solution that can detect allergens efficiently and provide users with safe meal options personalized to their specific dietary restrictions and preferences.

3.1.2 Existing System

In the existing scenario, food allergen detection is primarily a manual process that depends on the user's ability to read and interpret ingredient labels or consult online resources. This approach is error-prone and inefficient, as food labels may not always list hidden allergens or cross-contamination risks. There are some basic applications that display general allergen information, but most lack personalization, real-time detection, and intelligent recommendations. Additionally, these systems often do not maintain a user profile or record of allergens, forcing users to re-enter information repeatedly. The absence of automation and personalized feedback leads to poor user experience and limited reliability in allergen detection.

3.1.3 Proposed System

The proposed Food Allergen Detection and Safe Meals System aims to overcome the limitations of existing manual and static methods by providing an intelligent, web-based platform for allergen detection and safe meal recommendations. The system enables users to register with their allergen details, scan or enter food information, and instantly determine whether the food item is safe to consume. The backend, developed with Node.js and Express.js, analyzes the ingredient list against the user's allergen database stored in MongoDB, while the frontend, built using HTML, CSS, JavaScript, and Tailwind CSS, provides a simple, interactive user interface. The system also suggests **alternative safe meals** based on user preferences and cuisine type, promoting safe and enjoyable dining experiences.

Advantage:

- **Personalized Allergen Detection:** Automatically identifies allergens in food items based on user-specific allergy profiles.
- **Enhanced Accuracy and Safety:** Reduces human error in allergen identification by using automated comparison algorithms.
- **Real-Time Analysis:** Instantly analyzes scanned or entered food details and displays whether the food is safe or unsafe to consume.
- **Smart Meal Recommendations:** Suggests allergen-free meal alternatives based on cuisine preferences and dietary restrictions.
- **Scalability:** The system can easily accommodate a growing database of users and food items, ensuring future expansion without performance issues.

3.2 REQUIREMENT ANALYSIS

3.2.1 Functional Requirements

These are the functional and quality constraints that the system must satisfy to ensure reliability, user satisfaction, and data security. The extent to which these features are implemented depends on the project's scope and user needs:

- **User Authentication:** Provides secure login and registration for users to ensure that only authorized individuals can access personalized profiles and allergen data.
- **User Profile Management:** Users can manage their personal details, food preferences, and allergen information. The system maintains this data for personalized meal recommendations.
- **Image Upload and OCR Detection:** Allows users to upload images of food packages, labels, or menus. The system extracts text from the image using Optical Character Recognition (OCR) and processes it for allergen detection.
- **Allergen Detection and Analysis:** Compares extracted ingredient data with the allergen database to identify potential allergens and highlight unsafe components.
- **Safe Meal Recommendation:** Recommends safe meals and food items based on the user's allergen profile and selected cuisine preferences.
- **Multilingual Support:** Provides multilingual detection and translation for ingredient data (English, Tamil and Hindi), ensuring accessibility for diverse users and imported food products.
- **Search and Filtering:** Enables users to search for meals, ingredients, or allergens based on keywords or categories, improving the ease of use and accessibility of data.

- **Feedback and Rating System:** Allows users to rate meals and provide feedback on their safety and quality, helping improve allergen detection and reliability.
- **Data Storage and Management:** Centralized storage for user profiles, allergen data, and meal information using a secure database for fast access and easy updates.
- **Data Security and Backup:** Allows Implements data encryption and backups to ensure user privacy and prevent data loss.

3.2.2 Non-Functional Requirements:

These are the quality attributes and constraints that ensure the Food Allergen Detection and Safe Meals system performs efficiently, securely, and reliably under all conditions.

- **Performance:** The system should efficiently process image and text inputs to detect allergens in real time using optimized algorithms and API integrations. The response time for allergen detection and meal safety suggestions must remain under a few seconds, ensuring a seamless user experience.
- **Scalability:** The system should be designed to handle a growing number of users, meal entries, and allergen datasets without degrading performance. It must allow easy integration of new cuisines, food items, and additional allergen databases as the user base expands.
- **Availability:** The application must be available 24/7 with minimal downtime, ensuring users can access allergen detection and safe meal recommendations anytime, especially when dining out or traveling.
- **Reliability:** The system should consistently provide accurate allergen detection results and meal recommendations without failure. It must ensure data integrity during user interactions and prevent system crashes during

concurrent access.

- **Security:** Sensitive user data, including personal allergen profiles and login credentials, must be protected through encryption and secure authentication mechanisms. The system should employ role-based access control to prevent unauthorized modifications to the database.
- **Usability:** The interface must be user-friendly, multilingual, and intuitive, allowing users from various backgrounds to navigate easily. The allergen detection and safe meal recommendation process should require minimal steps and clear feedback.
- **Maintainability:** The system should have a modular and well-documented architecture, allowing easy maintenance, debugging, and updates. New features, APIs, or allergen datasets should be integrated without major changes to the core system.
- **Accuracy:** The allergen detection module should maintain high accuracy levels in identifying allergenic ingredients and providing safe meal suggestions. Continuous updates to the allergen database and model training should enhance precision over time.

3.2.3 Responsiveness:

The web application must be responsive across multiple devices, including desktops, tablets, and smartphones. It should automatically adjust layouts and input fields for different screen sizes to maintain accessibility and ease of use.

3.2.4 Hardware Specifications

- Processor (CPU) : Intel Core i5 or newer
- Hard disk : Minimum 500 GB
- RAM : Minimum 4 GB

3.2.5 Software Specifications

- Operating System : Window
- Tool : Visual Studio Code
- Language : Node.js
- Database : JSON
- Framework : Tailwind CSS, Mongoose, Express.js

CHAPTER 4

DESIGN

4.1 OVERALL DESIGN

The Food Allergen Detection and Safe Meals system is a web-based application designed to help users identify food allergens and choose safe meals based on their dietary restrictions. The system follows a client-server architecture, ensuring that all operations such as allergen detection, meal safety analysis, and user authentication are handled efficiently and securely across multiple devices, allowing users to access personalized meal recommendations anytime and anywhere.

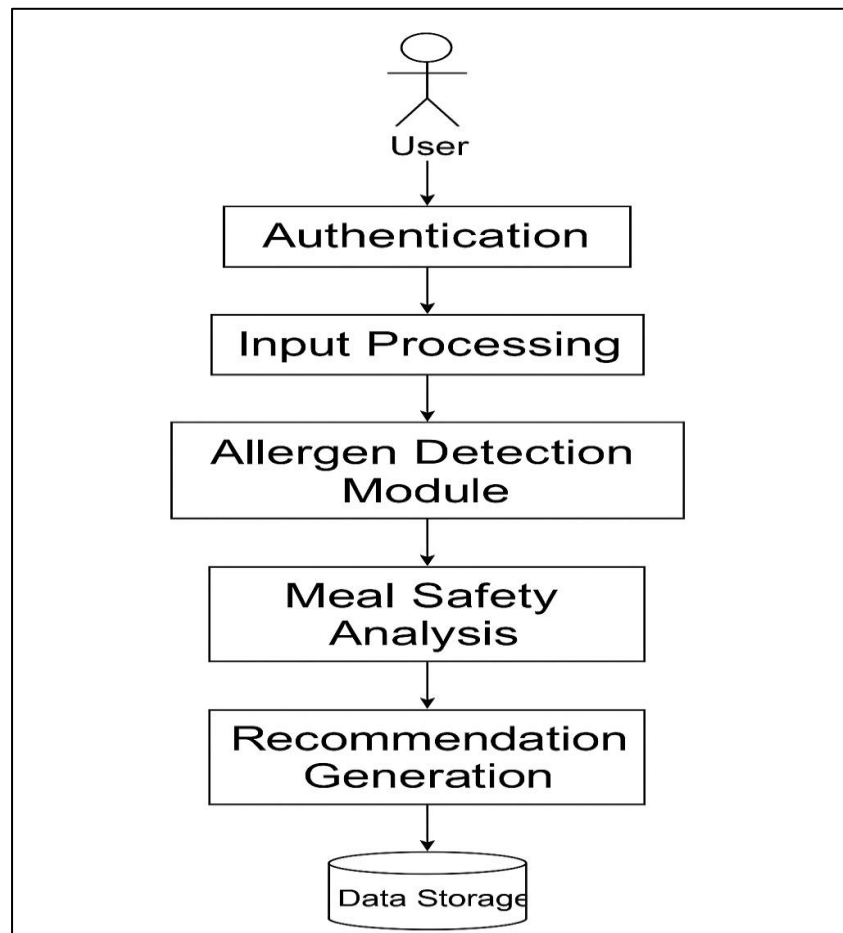


Fig 4.1 Proposed System Architecture.

4.2 UML DIAGRAMS

4.2.1 Work flow diagram

The workflow diagram represents the step-by-step process of how the Food Allergen Detection and Safe Meals system operates, from user interaction to generating meal safety results. It visually illustrates how data flows through different modules such as user authentication, allergen detection, meal safety analysis, and recommendation generation, ensuring a clear understanding of the overall system functionality and the sequence of operations.

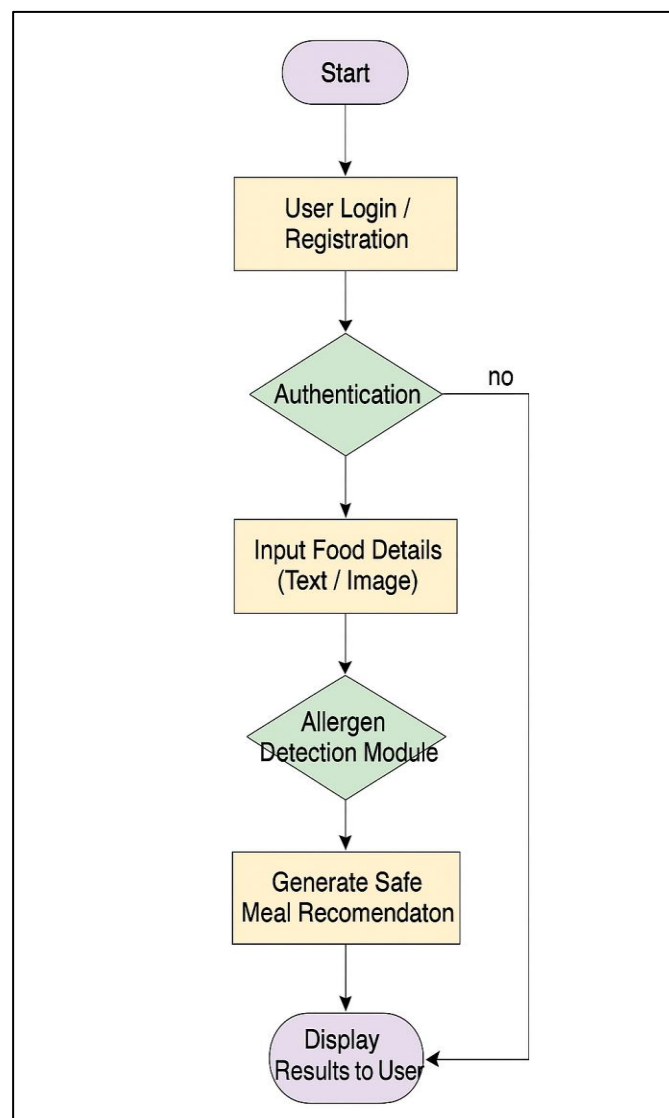


Fig 4.2 Work flow Diagram

4.2.2 Use Case Diagram

A Use Case Diagram for the Food Allergen Detection and Safe Meals System provides a graphical overview of the system's core functionalities, illustrating how users and administrators interact with it. It identifies the key actors, their goals, and the system's responses to their actions.

Use case: A use case represents a specific interaction between the user and the system, showing how users can perform tasks such as detecting allergens, viewing meal details, and submitting feedback. Each use case is depicted as a horizontal ellipse.

Actor: An actor can be a User, Admin, or an External API (e.g., allergen or food data source). These actors represent entities that interact with the system for various purposes — users detect allergens and rate meals, while admins manage meal data and user reports.

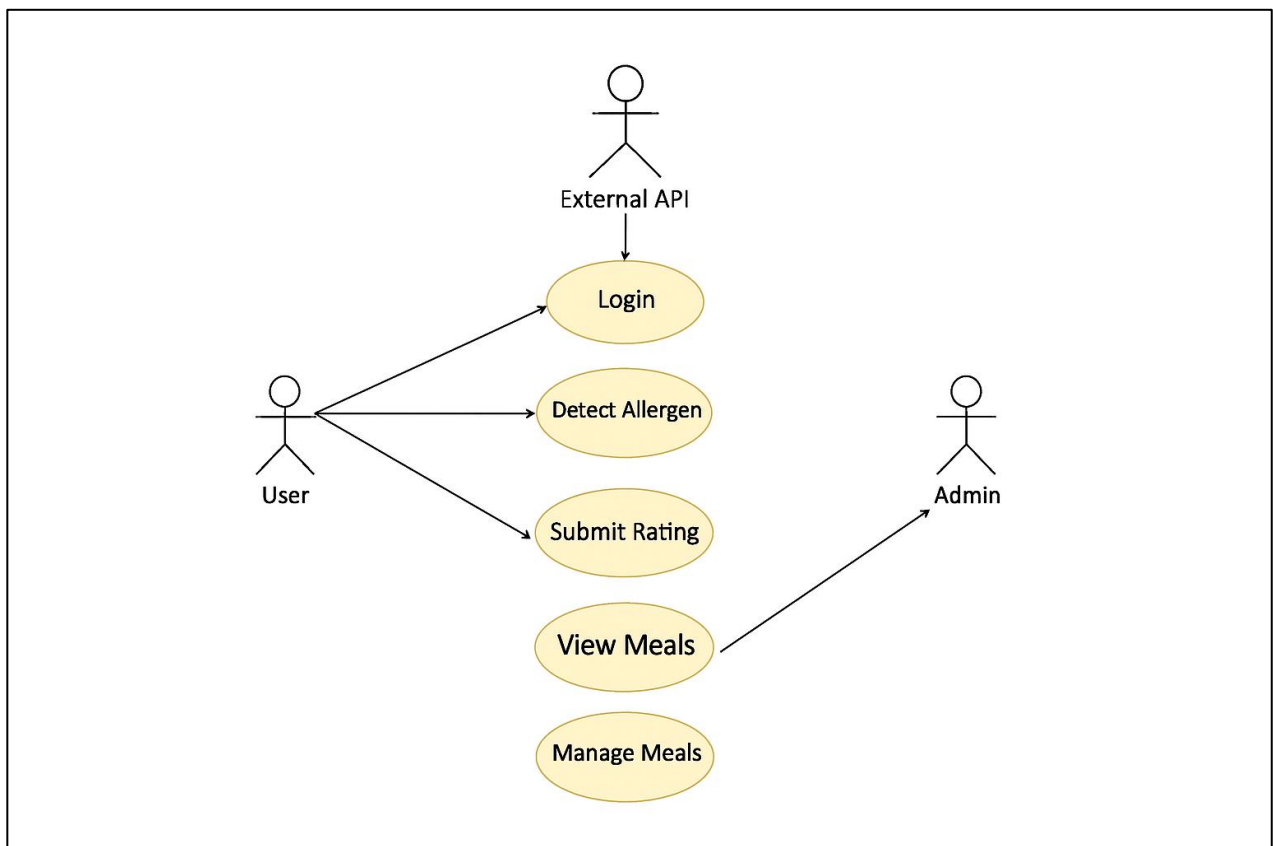


Fig 4.3 Use Case Diagram

4.2.3 Class Diagram

The Class Diagram for the Food Allergen Detection and Safe Meals System represents the structural design of the application, depicting the key classes, their attributes, and relationships. It provides a static view of the system and highlights how data and functionalities are organized and connected.

Each class in this project corresponds to a key component of the system such as users, meals, allergens, and feedback. The diagram helps developers and stakeholders understand the logical relationships and dependencies within the system, ensuring scalability and maintainability.

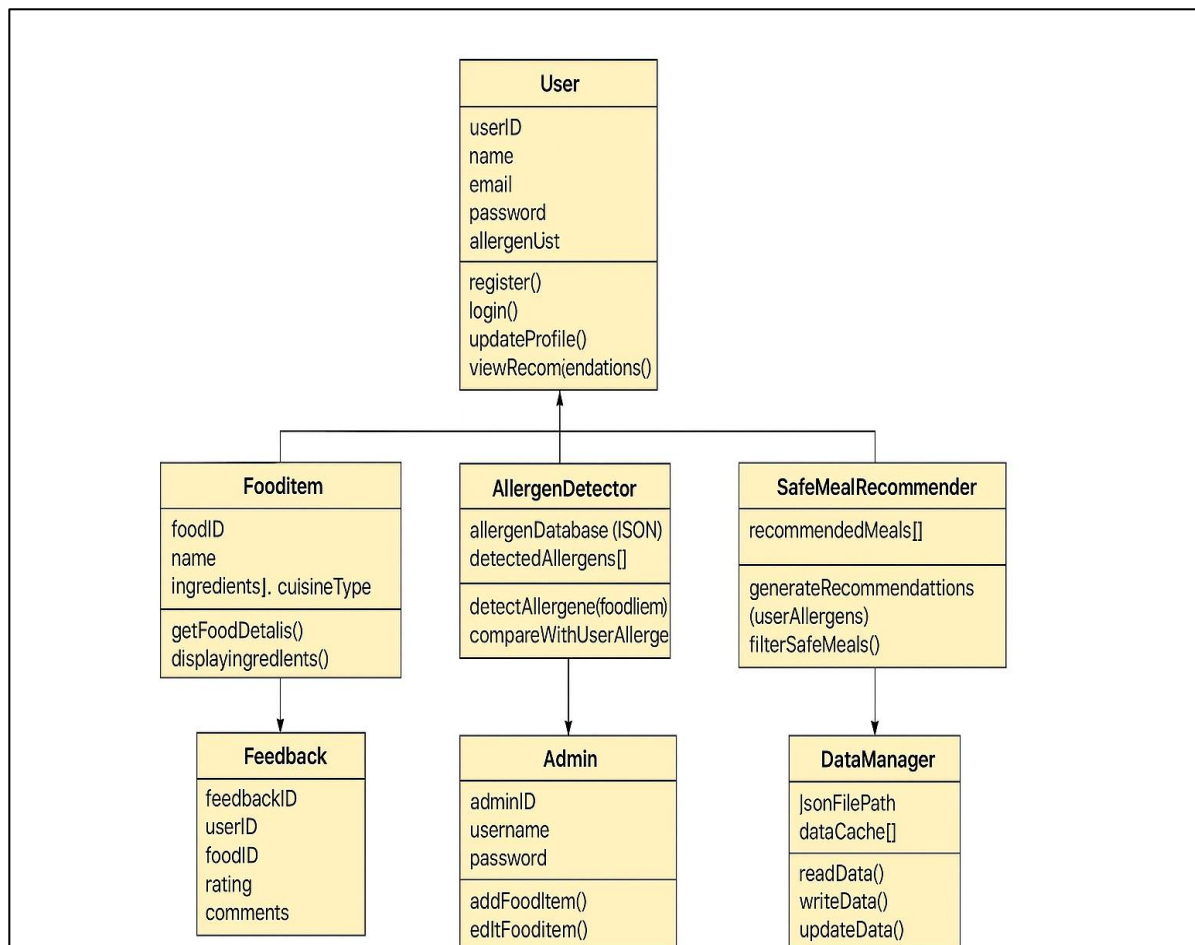


Fig 4.4 Class Diagram

4.2.4 Activity Diagram

An activity diagram represents the dynamic behavior of the Food Allergen Detection and Safe Meals system by illustrating the sequence of activities performed during various operations, such as user authentication, meal scanning, allergen detection, and displaying safe meal recommendations. It helps in visualizing the workflow and understanding how different processes are connected within the system. Activity diagrams are not only useful for visualizing the dynamic aspects of the system but also aid in constructing the executable system through forward and reverse engineering techniques. Although they resemble flowcharts, activity diagrams do not show message flows between activities; instead, they focus on the control flow and logical sequence of operations, including parallel, branched, concurrent, and sequential flows that occur within the application.

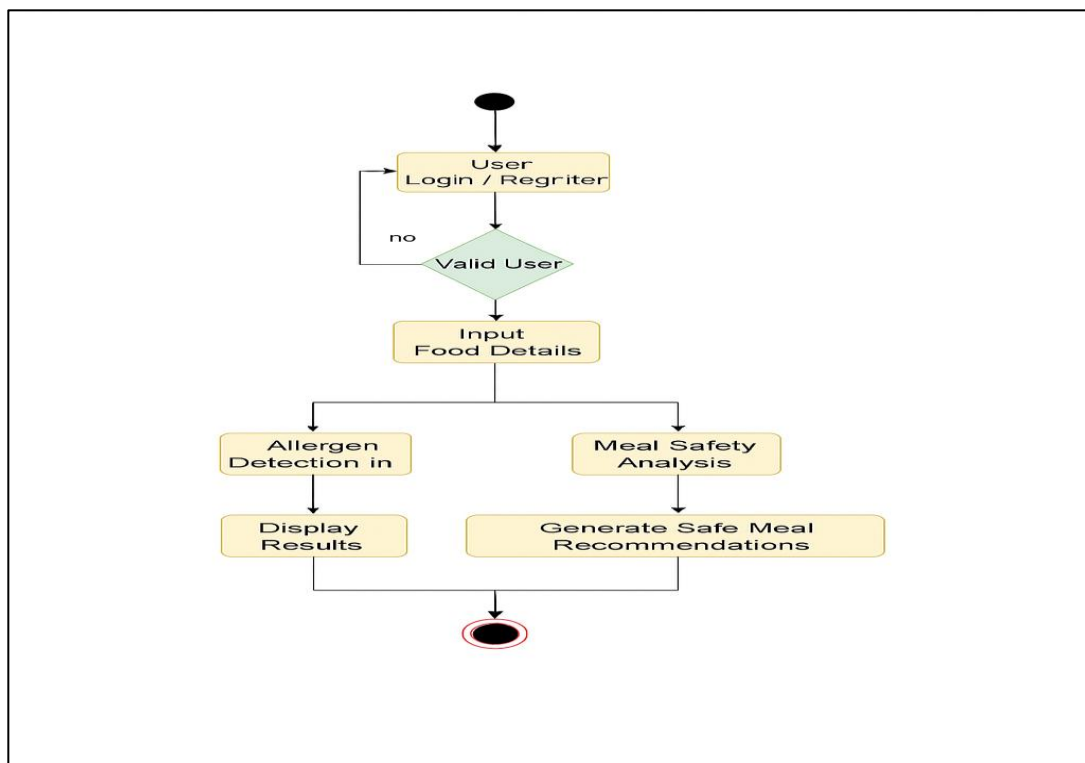


Fig 4.5 Activity Diagram

4.2.5 Sequence Diagram

A Sequence Diagram for the Food Allergen Detection and Safe Meals System visually represents the interaction between users, the system interface, and backend components in a time-ordered sequence. It demonstrates how data flows during key processes such as user authentication, food image upload, allergen detection, and meal safety recommendation. This diagram helps in understanding the dynamic behavior of the system, showing how the user's actions trigger sequential operations like input validation, allergen analysis using APIs, data retrieval from JSON, and safe meal suggestions. It effectively models the step-by-step communication among components, ensuring that each function performs in a logical and coordinated order for accurate allergen detection and user satisfaction.

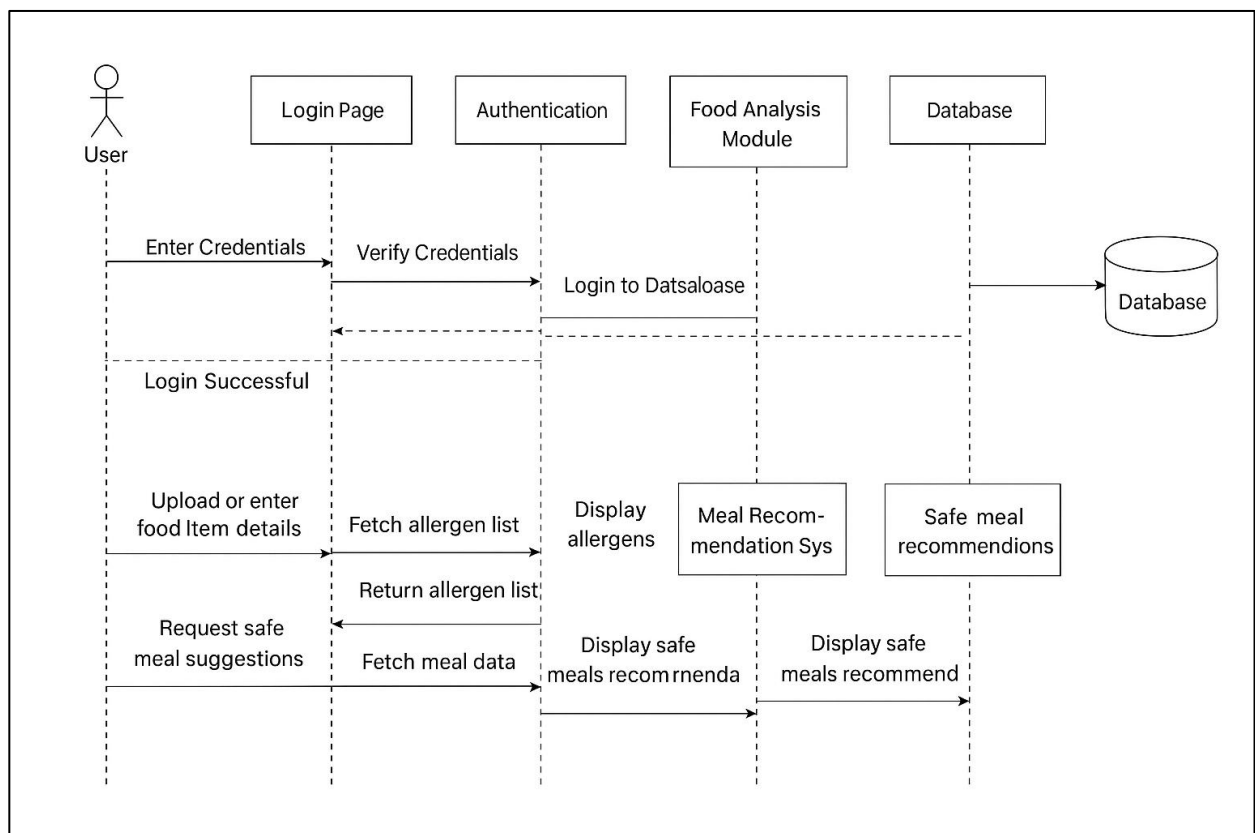


Fig 4.6 Sequence Diagram

CHAPTER 5

IMPLEMENTATION

5.1 MODULES

The Food Allergen Detection and Safe Meals System is designed to analyze food ingredients for allergen risks and recommend safe meal options to users. To achieve this, the system is divided into the following main modules:

- Food Analysis Module
- Meal Recommendation Module
- Database Management Module

5.2 MODULE DESCRIPTION

5.2.1 Food Analysis Module

Purpose:

- To detect allergen present in food ingredients using user-provided input or uploaded data.

Features:

- Users can upload food ingredient lists or enter food names manually.
- The system cross-verifies ingredients against the Food allergen detection and Safe meals database.
- Identifies and displays possible allergens like peanuts, gluten, dairy, etc.
- Provides visual alerts (e.g., “Contains Allergen” or “Safe to Eat”).
- Uses API integration or stored allergen datasets for real-time analysis.

5.2.3 Meal Recommendation Module

Purpose:

- To recommend meals that are safe based on the user's allergy profile.

Features:

- Retrieves user allergy data and filters out unsafe meals.
- Suggests safe alternatives for foods containing allergens.

5.2.4 Database Management Module:

Purpose:

- To handle all backend data operations for users, meals, and allergens.

Features:

- Maintains collections for user data, allergen information, and safe meal records.
- Performs CRUD operations (Create, Read, Update, Delete) on stored data.
- Provides efficient retrieval of allergen and meal data for faster recommendations.
- Ensures data integrity, scalability, and security across all modules.

CHAPTER 6

TESTING

6.1 TESTING AND VALIDATION

Testing and validation are critical to ensure that the Food Allergen Detection and Safe Meals System functions accurately, securely, and efficiently. Various testing methods were carried out to verify that all modules work as expected and meet the defined requirements.

Unit Testing:

- Each individual component of the system was tested separately to ensure proper functionality.
- **Backend:** Verified food analysis APIs, allergen detection logic, and database operations for storing user and meal data.
- Each individual component of the system was tested separately to ensure proper functionality.
- **Frontend:** Tested user interface components like login forms, allergen input sections, and meal suggestion displays.
- Ensured that all modules perform as expected in isolation before integration.

Integration Testing:

- Verified the smooth communication between frontend, backend, and the database.
- Ensured that user inputs (e.g., food ingredients) are properly analyzed by the backend and that results are accurately displayed on the frontend.

- Checked that data retrieval and updates between the allergen detection module and recommendation system work seamlessly.

Functional Testing:

- Tested whether the system performs all key operations as intended.
- Verified core functions like user login, allergen detection, safe meal recommendation, and profile management.
- Ensured accuracy in allergen identification and correctness of recommended meals according to stored user allergy data.

End-to-End (E2E) Testing:

- Simulated the complete user workflow from registration and login to allergen detection and safe meal recommendations.
- Validated that all user actions produce correct results without system errors or data loss.
- Confirmed that the system provides a smooth, error-free experience across all modules.

Performance Testing:

- Evaluates the system's behavior under different loads, particularly when multiple users scan food items or request allergen detection simultaneously.
- Ensures smooth performance during large data processing, such as analyzing ingredients or generating meal recommendations for many users at once.

Security Testing:

- Ensures that the system is protected against threats like unauthorized access, SQL injection, and data breaches.
- Verifies secure handling of sensitive user data such as personal information, allergy details, and login credentials.

User Acceptance Testing (UAT):

- Involves real users and administrators testing the system to confirm that it meets all usability and functional requirements.
- Validates that features like allergen detection, safe meal suggestions, and email notifications function accurately and intuitively.

Load Testing:

- Assesses the system's ability to manage multiple concurrent user performing allergen detection, scanning ingredients, and retrieving meal suggestions.
- Tools like Apache JMeter or k6 can simulate real-world where several users access the system simultaneously.

Response Time Testing:

- Measures how quickly the system responds to user actions such as scanning food, detecting allergens, or sending email notifications.
- Ensures optimal response time for a smooth and reliable user experience.

Security Testing

Authentication and Authorization:

- Ensures that sensitive operations (like managing allergen databases or updating meal records) are restricted to authorized admins only.
- Uses security tools such as OWASP ZAP or Burp Suite to detect vulnerabilities like SQL Injection, Cross-Site Scripting (XSS), and CSRF attacks.

Data Protection:

- Verifies that all sensitive data, including allergy details and personal health information, is properly encrypted during storage and transmission.
- Ensures compliance with security standards to prevent unauthorized data exposure.

User Acceptance Testing (UAT):

- Involves real users with different dietary profiles to test allergen detection accuracy and voice assistant functionality.
- Confirms that the system provides accurate results, easy navigation, and a user-friendly interface across devices.

Validation

Input Validation:

- Ensure that all input fields are validated both on the client and server sides.
- **Example:** Validate food names (non-empty, valid characters only) and allergy details (proper format, no duplicates).

- Check that only supported ingredients or meal names are accepted during allergen detection.

Role-Based Access Control:

- Ensure that only authorized users can access specific modules.
- **Admin:** Can manage allergen data, update safe meal lists, and modify system records.
- **User:** Can perform allergen detection, update personal allergy details, and view meal recommendations.
- Prevent unauthorized users from accessing or modifying backend data.

Data Integrity:

- Ensure data consistency across all modules and databases.
- Avoid duplicate entries of the same meal or allergen record.
- When a user deletes their account, all associated data such as allergy details and preferences should also be removed.
- Maintain accurate mapping between user profiles, detected allergens, and safe meal suggestion.

Token and Session Validation (JWT):

- Validate all user sessions using secure authentication tokens.
- **Example:** Each login session is verified using encrypted JSON Web Tokens (JWT).
- Automatically expire inactive sessions to enhance system security and prevent unauthorized access.

Database Validation:

- Verify data consistency and structure within the system database.
- Ensure no orphaned records exist (e.g., meals without allergen references or users without profiles).
- Validate that all stored data — including allergens, ingredients, and meal details — remain synchronized and accurate after each operation.

6.2 BUILD THE TEST PLAN

Table 6.1 Test case Design

Test Case ID	Test Scenario	Test Description	Input Data	Expected Output	Actual Result
TC001	User Registration	Verify that a new user can register successfully with valid details.	Name, Email, Allergies	Account created successfully	Pass
TC002	User Login	Validate login with correct credentials.	Email, Password	Redirect to user dashboard	Pass
TC003	Invalid Login	Check login attempt with incorrect credentials.	Wrong Email/Password	Display “Invalid credentials” error	Pass
TC004	Allergen Detection	Verify system correctly identifies allergens from given ingredients.	Food Name: “Peanut Butter”	Displays “Contains Allergens: Peanut”	Pass
TC005	Safe Meal Recommendation	Ensure system recommends meals safe for user’s allergies.	Allergy: “Gluten”	Suggests gluten-free meals only	Pass
TC006	Profile Update	Verify user can update allergy information.	Add new allergy: “Dairy”	Profile updated successfully	Pass
TC007	Admin Access Control	Check that only admin can manage allergen data and meal records.	Admin credentials	Admin panel accessible	Pass
TC008	Unauthorized Access	Ensure restricted modules are inaccessible to normal users.	User login	Access denied message	Pass
TC009	Session	Verify	Idle user session	Auto logout	Pass

	Validation	JWT/session token expiration after inactivity.		after timeout	
TC010	Database Consistency	Check that all updates reflect correctly in the database.	Add/Delete allergen record	Database updated accurately	Pass

CHAPTER 7

RESULT AND DISCUSSION

7.1 RESULTS

The Food Allergen Detection and Safe Meals System was thoroughly tested at unit, integration, functional, and end-to-end levels, and all modules performed successfully. The authentication process worked accurately, allowing users to register, log in securely, and access personalized dashboards, while enforcing proper role-based access between admin and user roles. The food analysis module correctly identified allergens from user-entered ingredients, displaying clear warnings for unsafe items and validating all inputs effectively. The meal recommendation module generated accurate, allergen-free meal suggestions based on each user's stored allergy profile, ensuring relevant and safe results. User profile updates were reflected instantly, maintaining data integrity across all modules. The database operations, including insert, update, and delete, executed smoothly without redundancy or loss, ensuring consistent and reliable data management. Overall, the system demonstrated strong reliability, accuracy, and security in detecting allergens and recommending safe meals, meeting all functional and performance requirements effectively.

7.2 DISCUSSION

The Food Allergen Detection and Safe Meals System underwent multiple stages of testing, including unit, functional, and system testing, and all modules performed effectively. The user authentication and registration modules functioned accurately, ensuring secure login and proper role-based access control for both users and admins. The allergen detection feature

successfully analyzed food ingredients and identified potential allergens based on stored user data, with error-free validation and accurate warnings for unsafe food items. The safe meal recommendation module generated appropriate allergen-free meal suggestions, confirming the system’s ability to tailor results according to individual dietary profiles. Database operations such as adding, updating, and deleting records worked seamlessly, maintaining integrity and preventing data duplication. Overall, the system met all functional and performance requirements, proving to be reliable, secure, and efficient in ensuring safe meal recommendations for allergy-prone users.

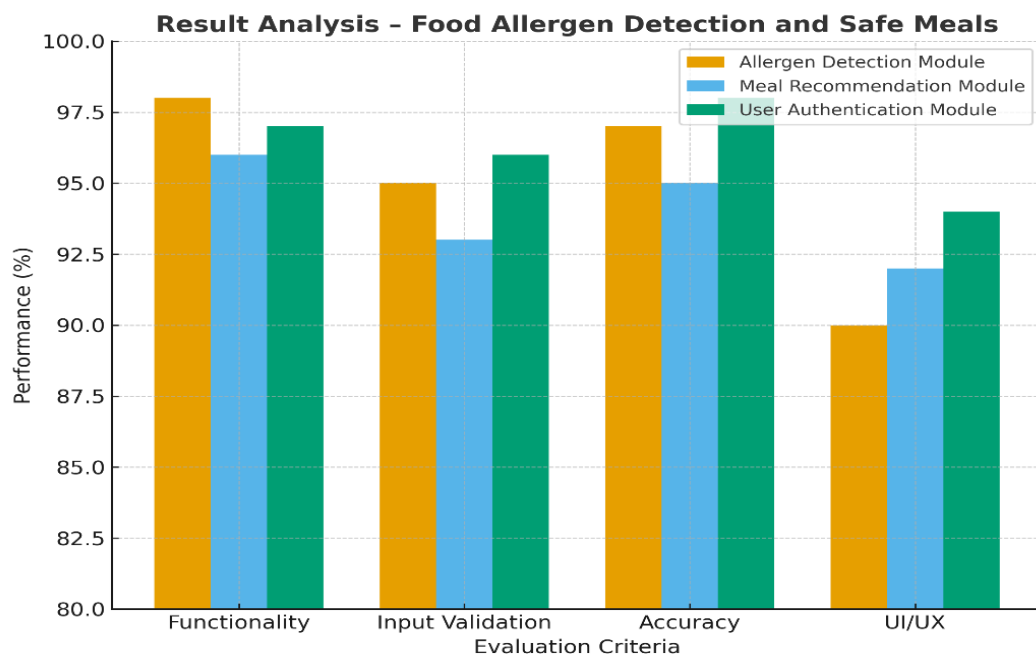


Fig 7.1 Result Analysis

CHAPTER 8

USER MANUAL

Installing Visual Studio Code:

- Step 1:** Visit the official Visual Studio Code downloads page. Go to the following link: code.visualstudio.com/Download.
- Step 2:** Select the appropriate version for your operating system (Windows, macOS or Linux)
- Step 3:** Download the installer and run the setup.
- Step 4:** Follow the installation prompts, accepting the license agreement and default settings
- Step 5:** Once the installation is complete, launch Visual Studio Code.
- Step 6:** Optionally, install extensions for Python and Jupyter within VS Code.

Executing Modules in VS Code:

- Step 1:** Open Visual Studio Code and navigate to the relevant project folder containing your Project directory.
- Step 2:** Select the file you wish to run and ensure that the correct server is selected in the bottom left corner of the window. You can change the server port, if necessary, by clicking on it and choosing the appropriate one from the list.
- Step 3:** Before executing the code, check the code for any syntax errors or issues. You can use the built-in feature to identify and fix any problems.

Step 4: Check for the errors while compiling the program, if no error is found then run a program.

Step 5: Observe the output in the integrated terminal to verify that the program executed successfully.

Installing Node Js:

Step 1: Go to the official website of Node js and download the current version of node

Step 2: Select the appropriate version for your operating system
(Windows, macOS or Linux)

Step 3: Download the installer and run the setup.

Step 4: Follow the installation prompts, accepting the license agreement and default settings

Step 5: Once the installation is complete, launch Node server .

Installation Guide :

Install the necessary packages needed for our project

Express JS: npm install express

Mongoose: npm install mongoose

Axios: npm install axios

Run command: node server.js

CHAPTER 9

CONCLUSION

In Conclusion, The testing phase of the Food Allergen Detection and Safe Meals System confirmed its reliability, accuracy, and functionality across all core modules, including user authentication, allergen detection, and meal recommendation. All major features performed as expected, allowing users to securely register, detect allergens in food items, and receive personalized safe meal suggestions. The admin functions, such as updating allergen data and managing safe meal records, also operated efficiently without errors.

Enhanced input validation and robust error handling contributed to smoother system performance by accurately flagging invalid inputs and preventing data inconsistencies. The system demonstrated strong scalability and performance during testing with larger datasets, ensuring its capability to handle multiple user requests simultaneously. Minor improvements in the user interface and experience were implemented to make the platform more intuitive, responsive, and user-friendly.

In conclusion, the system proved to be stable, secure, and ready for deployment, with optimized accuracy, scalability, and performance. It effectively supports allergen detection and safe meal recommendations, providing a reliable digital solution for users with dietary restrictions and food allergies.

CHAPTER 10

FUTURE ENHANCEMENT

To further enhance the Food Allergen Detection and Safe Meals System, several advanced features can be integrated to improve efficiency and user experience. A food scanner module can be added to instantly analyze ingredients from food labels or menus using image recognition, making allergen detection faster and more accurate. An automated email notification system can inform users about allergen risks, safe meal recommendations, and newly added food items, ensuring users stay updated in real time.

Integrating a voice assistant would provide hands-free interaction, allowing users to check allergen information or request meal suggestions through simple voice commands, improving accessibility for all age groups. Additionally, enhanced analytics can help track user preferences, allergen patterns, and dietary trends for personalized health insights.

In the future, expanding the system with AI-based allergen prediction and cloud synchronization will support large-scale data processing and multi-device access. These enhancements will make the system smarter, more interactive, and user-friendly, ultimately helping users make safer and healthier food choices with confidence.

APPENDIX-I
BASE PAPER

Food Recommendation with Balanced Nutrition Using Context-Aware Knowledge-Base

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Abstract—The food recommender system plays a crucial role in providing solutions for the community, as there are many food choices available in society. However, the lack of attention to the nutritional value of the foods we consume can lead to obesity. The increasing prevalence of obesity worldwide is an urgent problem that needs to be addressed. Obesity is a serious issue as it can cause various dangerous diseases such as diabetes, atherosclerosis, cholesterol, and even cancer. This necessitates the need for a food recommender system that considers not only user preferences but also nutritional needs. This research aims to develop a food recommender system for various age groups that consider the nutritional needs, diseases, and allergies that the user is suffering from. Therefore, we propose a knowledge-based context-aware approach to building a food recommender system. The proposed recommender system utilizes a chatbot to facilitate user's interactions with the system. The system leverages information from the internal context about user characteristics such as age, gender, height, weight, activity level, and medical history. Additionally, the information on external context relates to food types and recent weather. There are 11 types of food that are recommended. Each recommended food is tailored to user preferences and nutritional needs. Testing results of our proposed system show high accuracy, with an F1-Score reaching 0.97.

Keywords—recommender system, ontology, knowledge-based recommender system, knowledge-based context-aware system, Semantic Web Rule Language

I. INTRODUCTION

The rapid development of Information Technology has significantly influenced our lives today. Often, due to various daily responsibilities, consuming healthy and balanced nutrition becomes the last option for us. We tend to consume fast food where nutritional value is not prioritized, which can lead to obesity. In 2016, there were 650 million obese adults aged 18 and above, and 41 million obese children [1]. Obesity has caused various serious health problems that exist today [2]. Blocked blood vessels, stroke, diabetes mellitus, hypertension, and even cancer are some dangerous diseases caused by obesity [3]. With the increasing number of obesity cases worldwide in the past decade, issues related to nutritional value must be seriously considered [4]. It is crucial to understand a healthy lifestyle [5] to avoid the dangers of obesity. However, adopting a healthy lifestyle is not a simple task because many of us lack the necessary skills and knowledge to effectively implement it, thus making a positive impact on our lives [6].

In this research, we develop a recommender system that assists users in recommending nutritionally customized foods while also considering suitable options for users with allergies, diabetes, and cholesterol concerns. Previous studies have built food recommender systems using various approaches, including collaborative filtering [7], content-

based [8], [9], knowledge-based [10], [11], and even hybrid models [2]. Vivek et al. [12] developed a recipe recommender system using a collaborative filtering approach but faced difficulties with the dataset processing, as it relied on user ratings to provide recommendations. Almeida [8] built a recommender system using a content-based approach, but encountered limitations in terms of content availability, as the dataset used was not sufficiently informative. Ahuja et al. [13] developed a recommender system by combining two approaches: collaborative filtering, which analyses food preferences suitable for users, and content-based analysis, which examines food characteristics and recommends similar items to the user's preferred choices. Meanwhile, Jung et al. [2] created a recommender system using a knowledge-based context-aware approach along with collaborative filtering. However, the nutritional standards were specifically designed for Korean food, and the recommender system targeted obesity prevention in young individuals, despite obesity being prevalent across all age groups. Therefore, there is a need for a recommender system capable of suggesting Indonesian foods based on appropriate nutritional standards for the Indonesian population. The built food recommender system should be usable by users of all age groups and aim to provide food recommendations aligned with their nutritional needs. The knowledge-based approach utilizes existing knowledge to reason about recommended products according to user needs [10]. This approach can address issues related to cold start for new users and data sparsity.

In this research, we propose a knowledge-based context-aware approach to gather information from both internal context and external contexts. We employ the knowledge-based context-aware approach to ensure that the food recommendations obtained by the user are more accurate.

II. RESEARCH METHODS

The research began by collecting food data from the Indonesian Food Composition Table in 2017. The ontology is created using a context-aware system, and we use Protege as the tool for composing the ontology. An ontology is a conceptual model used to describe correlations and fundamental ideas related to a specific domain [2], [14]. Once the ontology is completed, a chatbot is developed using the Telegram platform, and we use Python programming language. Subsequently, food recommendations are provided to users, followed by calculating accuracy using precision, recall, and F1-score.

A. Foods Data

The data used is food data from The Indonesian Food Composition Table in 2017. We utilized 11 types of food, including fruits, cereals, eggs, oils, seeds and beans,

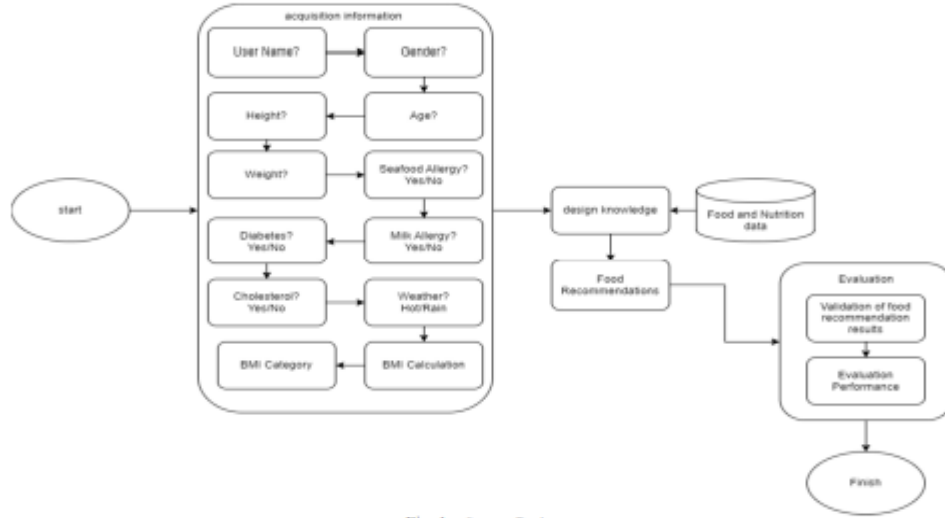


Fig. 1. System Design

vegetables, tubers, red meats and poultry, fish, shellfish and shrimp, milk and processed, and sugar syrups and confectionery. For each food item, we extracted nutritional values such as water, energy, protein, fiber, fat, carbohydrates, phosphorus, calcium, and vitamin C, as shown in TABLE I.

B. Knowledge-Based Context-Aware Recommender System

The knowledge-based recommender system is a recommender system that utilizes the knowledge possessed by the system, which is obtained from domain experts. Knowledge-based utilizes user preference-related knowledge, item-related knowledge, and knowledge from specialists to provide recommendations that are tailored to the user's needs [15], [16].

The knowledge-based context-aware approach is used to manage real-world context information and connect it when establishing connections between users and relevant services based on suitable knowledge [1]. Contextual data is classified into individual data, health data, Body Mass Index (BMI) categories, nutritional requirements, food types, and weather data.

In this phase, we build an ontology using contextual information. The contextual information we utilize comes from internal context such as individual data, health data, BMI category, and nutritional requirements, as well as external context, where we utilize food type data and weather data.

C. Chatbot

Chatbot is designed to respond to conversation-like messages from users by directly providing automated answers based on the given input [17]. Initially, chatbots were created to mimic human conversation and provide entertainment to users [18]. In this stage, we develop a chatbot based on the Telegram platform. The chatbot asks the user several questions to gather information. The questions include the user's name, age, gender, height, weight, allergies, medical conditions, activity level, and even the current weather conditions. After that, the system calculates the user's BMI and categorizes them into BMI categories as shown in TABLE II. The BMI value is obtained by

TABLE II. BMI categories based on WHO Global and Asia Pacific [21].

	WHO (BMI)	Asia-Pacific (BMI)
Underweight	≤ 18.5	≤ 18.5
Normal	18.5 - 24.9	18.5 - 22.9
Overweight	25 - 29.9	23 - 24.9
Obesity	≥ 30	≥ 25

calculating the user's height and weight using formula (1), as follows,

$$BMI = \frac{Weight(kg)}{Height(cm) \times Height(cm)} \quad (1)$$

Then, the chatbot provides food recommendations that align with the user's nutritional needs, considering their allergies and medical conditions.

D. Evaluation

In this stage, after obtaining food recommendations, the recommendations are validated by experts. Subsequently, accuracy is calculated based on the validation results, using precision, recall, and F1-score calculations. Precision, recall, and F1-score are calculated using formulas (2), (3), and (4).

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \quad (2)$$

$$Recall = \frac{True\ Positives}{False\ Negatives + True\ Positives} \quad (3)$$

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

The True Positive value represents the total number of food recommendations that are recommended to the user and are also validated by the expert. The False Positive value represents the number of food items recommended by the system but not validated by the expert. The False Negative value represents the number of food items that are not recommended by the system but should have been recommended according to the expert's validation.

TABLE I. Food data from The Indonesian Food Composition Table in 2017

Food Name	Food Type	Water	Energy	Protein	Fiber	Fat	Carbohydrates	Phosphorus	Calcium	Vitamin C
Apple	fruits	84.1	58	0.3	2.6	0.4	14.9	10	6	5
Shredded Beef	red meats and poultry	24.8	385	14.6	2.5	16.1	38.6	136	165	0
Rice	cereals	56.7	180	3	0.2	0.3	39.8	27	25	0

III. RESULT AND DISCUSSION

A. System Design

In Fig. 1, the system workflow consists of several stages. First, information related to internal and external context is collected to build the food recommender system, such as age, weight, height, activity level, weather conditions, etc. Second, knowledge is constructed by creating an ontology and SWRL (Semantic Web Rule Language) rules. Third, food recommendations are generated based on the user's nutritional needs and adjusted to the weather conditions at the user's location. Fourth, the evaluation process is conducted, divided into two stages: validation of the food recommendation results by an expert, followed by performance evaluation to assess the accuracy of the developed system.

B. Ontology Design

In Fig. 2, an ontology is constructed. There are two main classes in this ontology: external context and internal context. The external context class has two subclasses: weather with subclasses rain and hot, and food type with 10 subclasses: fruits and processed fruits, meat, poultry and processed poultry, fish, shellfish, shrimp and processed shrimp, starchy tubers and processed starchy tubers, vegetables and processed vegetables, nuts, seeds, beans and processed beans,

TABLE III. Object Property Used in Ontology

Properties Name	Domain	Range
hasFruits	FoodType	fruits
hasSugarandConfectionery	FoodType	SugarandConfectionery
hasFishShellfishandShrimp	FoodType	FishShellfishandShrimp
hasSeedsandBeans	FoodType	SeedsandBeans
hasOils	FoodType	Oils
hasRedMeatandPoultry	FoodType	redMeatandPoultry
hasVegetables	FoodType	vegetables
hasCereals	FoodType	cereals
hasMilkandProcessed	FoodType	milkandprocessed
hasEggs	FoodType	eggs
hasTubers	FoodType	tubers
hasType	food	FoodType
hasNeeds	dataperson	nutritionNeeds

sugar, syrup, and confections, cereals and processed cereals, and fats and oils. Moving on to the internal context, there are several subclasses: user preferences, personal data with

subclasses of activity data, health data, BMI category, and nutritional needs.

Object properties can be seen in TABLE III. Each property has its own task. For example, the object property **hasFruits** is responsible for expressing the relationship between the **fruits** class and **foodType**.

C. Chatbot

The initial stage of creating a chatbot starts with creating SWRL (Semantic Web Rule Language) rules. SWRL rules are used to obtain food recommendations from the previously built ontology. Several rules are created to obtain food recommendations that align with the user's nutritional needs. These rules include BMI classification, determining the user's calorie needs, user's activity level, and food selection. In this study, food recommendations are provided for four BMI categories mentioned in TABLE II. However, for users with a BMI above 23, the system categorizes them as obese. The BMI value stored in the data property **hasBMI**. The rule used to calculate the BMI value and determine the BMI category can be seen in Fig. 3.

```

person(?p), hasWeight(?p, ?bmi), hasHeight(?p, ?tinggi) ,
multiply(?bmi, ?bmi, ?bmi2), multiply(?tinggi, ?tinggi, ?tinggi2),
divide(?bmi2, ?tinggi2, ?bmi3) -> hasBMI(?p, ?bmi3)

person(?p), hasBMI(?p, ?bmi), greaterThanOrEqual(?bmi, 18.5), lessThanOrEqual(?bmi, 22.9) ->
normal(?p)

```

Fig. 3. BMI Category Rule

The calorie requirements for each user are determined according to the regulations set by the Ministry of Health of the Republic of Indonesia. The calorie needs of users are already specified based on their gender and age. However, for children under the age of 10, the calorie requirement is the same for both males and females. The calorie requirement for each user stored in the data property **hasEnergiNeed**. For example, the calorie requirement for a 9 years old child can be seen in Fig. 4.

```

person(?p), hasBMI(?p, ?bmi), hasAge(?p, ?a), greaterThanOrEqual(?a, 7), lessThanOrEqual(?a, 9) -> hasEnergiNeed(?p, 1650)

```

Fig. 4. Calory Requirement for 9 years old children

The next rule for male users aged 45 years and for female users aged 32 years, can be seen in Fig. 5.

```

person(?p), hasBMI(?p, ?bmi), hasAge(?p, ?a),
hasGender(?p, "pria"), greaterThanOrEqual(?a, 30), lessThanOrEqual(?a, 49) ->
hasEnergiNeed(?p, 2550).

person(?p), hasBMI(?p, ?bmi), hasAge(?p, ?a), hasGender(?p, "wanita"),
greaterThanOrEqual(?a, 30), lessThanOrEqual(?a, 49) -> hasEnergiNeed(?p, 2150)

```

Fig. 5. Calory Requirement Rule for male and female



```
person(?p).hasBMI(?p, ?bmi).hasAge(?p,
?a).hasGender(?p, ?g).hasActivity(?p, ?aktif).greaterThanOrEqualTo(?a, 16).lessThanOrEqualTo(?a,
16).multiply(?caloriActivity, 2550, 1.7)-> hasCaloriActivity(?p, ?caloriActivity).
```

[Return to top](#)

Food recommendations are tailored to each type. The type of fruit and process can be accessed using the object property `hasFruits`. Sugar and confectionery can be accessed using the object property `hasSugarandConfectionery`. Next, fish, shellfish, shrimp, and processed seafood can be accessed using the object property `hasFishShellfishandShrimp`. Then, nuts, seeds, beans, and processed versions can be accessed using the object property `hasSeedsandBeans`. Next, red meat, poultry, and processed versions can be accessed using the object property `hasRedMeatandPoultry`. Vegetables and processed versions can be accessed using the object property `hasVegetables`. Cereals and processed versions can be accessed using the object property `hasCereals`.

```

person(?p), obesity(?p), fruits(?fruit) -> hasFruits(?p, ?fruit)
person(?p), obesity(?p), vegetables(?vegetable) -> hasVegetables(?p, ?vegetable)
person(?p), obesity(?p), FishSteefishandShrimp(?fish) -> hasFishSteefishandShrimp(?p, ?fish)
person(?p), obesity(?p), cereals(?cereal) -> hasCereals(?p, ?cereal)
person(?p), obesity(?p), milkandprocessed(?milk) -> hasMilkandProcessed(?p, ?milk)
person(?p), obesity(?p), SugarsyrupandConfectionery(?sugar) ->
hasSugarsandConfectionery(?p, ?sugar)
person(?p), obesity(?p), tubers(?tuber) -> hasTubers(?p, ?tuber)
person(?p), obesity(?p), eggs(?egg) -> hasEggs(?p, ?egg)
person(?p), obesity(?p), SeedsandBeans(?beans) -> hasSeedsandBeans(?p, ?beans)
person(?p), obesity(?p), redMeatandPoultry(?meat) -> hasRedMeatandPoultry(?p, ?meat)
person(?p), obesity(?p), OilandFat -> hasOils(?p, ?oil)

```

Fig. 8 shows an example of interaction between the user and the chatbot. First, the user sends the "start" command, and

TABLE IV. User Data for Evaluation

Name	Gender	Age	Height	Weight	Seafood Allergy	Milk Allergy	Diabetes	Cholesterol	Kategori BMI
Hary	Male	45	159	57	No	No	Yes	No	Normal
Mark	Male	9	130	30	No	Yes	No	No	Underweight
Celine	Female	17	150	48	No	No	No	No	Normal
Justin	Male	70	168	64	No	No	Yes	Yes	Normal
Winney	Female	34	168	70	Yes	No	No	Yes	Obesity
Line	Female	88	154	48	No	No	Yes	Yes	Normal
Sindy	Female	68	157	57	No	Yes	Yes	Yes	Obesity
Tom	Male	26	169	60	Yes	No	No	Yes	Normal
Franz	Male	10	120	17	Yes	Yes	No	No	Underweight
Pristin	Female	12	132	45	No	No	Yes	Yes	Obesity

the system responds with a greeting. Then, the system asks for user preference with questions related to the user's name, gender, age, weight, height, activity level, allergies, diseases, and weather. After the user answers all the questions correctly, the system sends a message saying, "Mohon ditunggu (Please wait)", and for Fig. 9 system continues by sending a message containing the user's data and food recommendations. The proposed system will tailor the food recommendations to the user's needs. As shown in Fig. 9, the system can recommend foods that are appropriate for the user's needs. In the user's data, the user has a seafood allergy, and in the recommended food results, the system does not recommend seafood-based dishes. Since the user does not have a dairy allergy, diabetes, or cholesterol, the system still recommends foods containing milk and sugar, as well as fried foods.

Halof! Selamat Datang Di Bot Rekomendasi Makanan

Silakan masukkan nama:

Pilih jenis kelamin: ☐ ☐

Masukkan umur (dalam tahun):

Masukkan tinggi badan (dalam cm):

Masukkan berat badan (dalam kg):

Apakah anda memiliki alergi Seafood? ☐ ☐

Apakah anda memiliki alergi susu? ☐ ☐

Apakah anda menderita penyakit diabetes? ☐ ☐

Apakah anda menderita penyakit kolesterol? ☐ ☐

Bagaimana Cuaca hari ini? ☐ ☐

Pilih Level Aktivitas Fisik: ☐ ☐ ☐ ☐

Mohon ditunggu

Fig. 8. User Interaction on Chatbot (1)

System display user data

System recommends food.

System provides warning about the foods

REKOMENDASI MAKANAN:

BUAH: Buah dan Olahan: [Mangga, jeruk, apel, pisang]

SEREAL: Gerealita dan Olahan: [roti, pasta]

SAYUR: Sayuran dan Olahan: [bayam, brokoli]

DAGING: Daging, Unggas dan Olahan: [ayam, daging sapi]

GULA: Gula dan Olahan: [kacang, susu]

TELUR: Telur dan Olahan: [telur, telur ayam]

LEMAK: Lemak: [minyak, mentega]

KETERANGAN: Untuk pengguna yang mengalami alergi, jika ingin mengonsumsi makanan yang disarankan dengan cara di rebus atau dikukus, agar nutrisi dari makanannya lebih banyak didapatkan.

Fig. 9. User Interaction on Chatbot (2)

D. Evaluation

Validation of the food recommendations by experts was conducted on 10 users, as shown in TABLE IV. The recommended food for each user amounted to 20 food items. From the validated food recommendations by the experts, 3 menu items were not recommended, and 7 food items were not recommended by the system.

The performance evaluation of the system is calculating the validation results from the experts using formulas (2), (3), and (4).

$$\text{Precision} = \frac{200}{200 + 3} = 0.99$$

$$\text{Recall} = \frac{200}{7 + 200} = 0.97$$

$$F1 - \text{score} = 2 \times \frac{0.99 \times 0.97}{0.99 + 0.97} = 0.98$$

IV. CONCLUSION

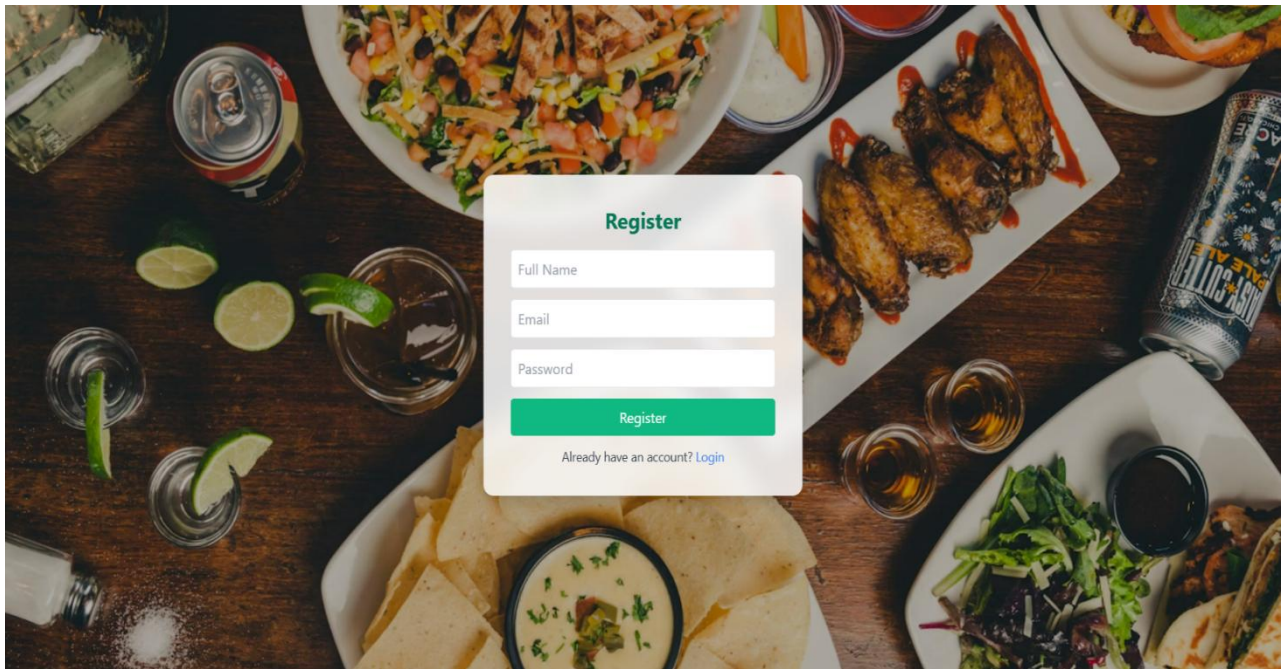
The knowledge-based context-aware recommender system has proven to successfully recommend food that aligns with user preferences while also meeting their nutritional needs. After the system was validated by experts and accuracy calculations were performed, the precision value obtained was 0.99, the recall value was 0.97, and the F1-score yielded a result of 0.98. From these accuracy calculations, it can be concluded that the food recommender system, utilizing the knowledge-based context-aware approach, has been successfully developed for users in the underweight, normal, and obese BMI categories. Overall, this research provides strong evidence that the use of a knowledge-based, context-aware approach achieves high-accuracy results. However, the food data used was specific to Indonesian cuisine. Therefore, the recommendation for future research is to use food data from other countries, while also adjusting the nutritional requirements for each user. This would allow the food recommendation system to be utilized by users from various countries, providing personalized recommendations that cater to their individual dietary needs.

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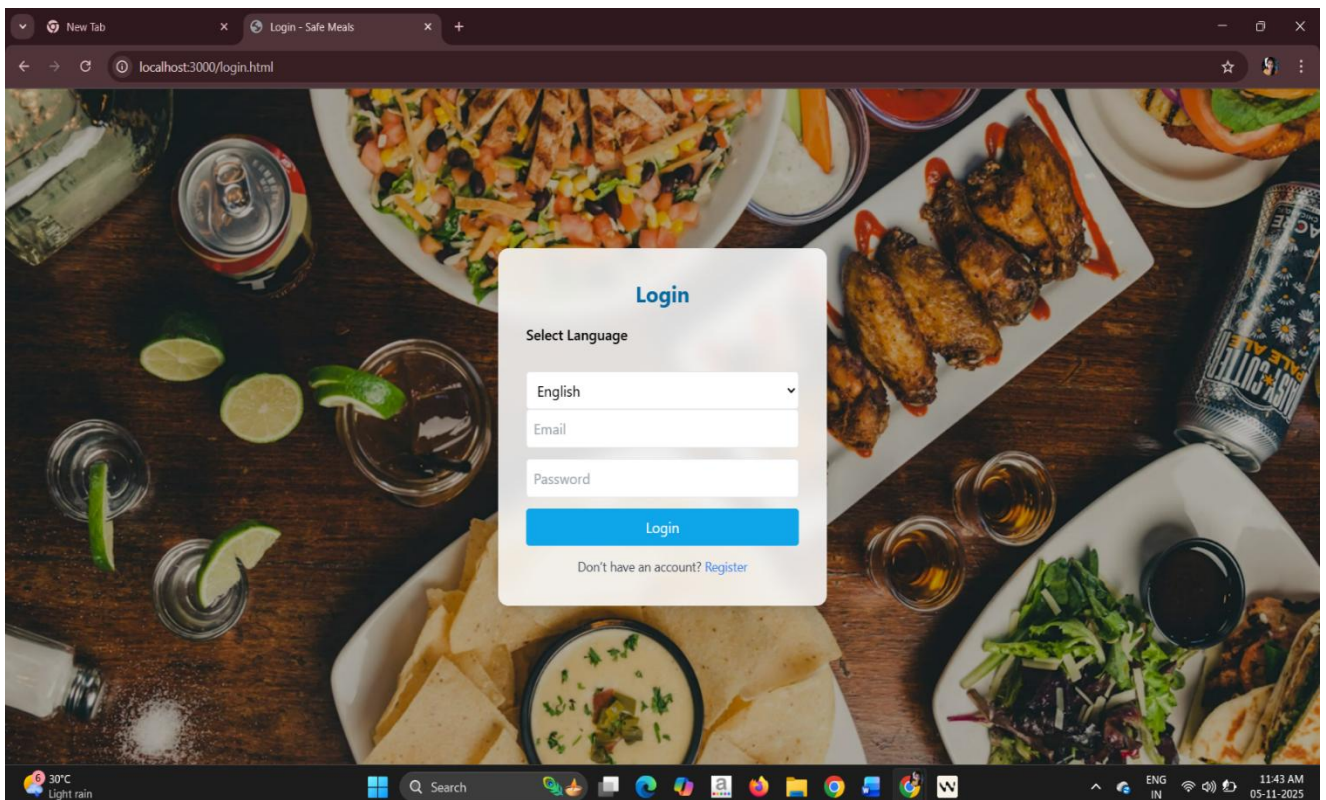
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APPENDIX-II
SCREENSHOTS

Registration Page:



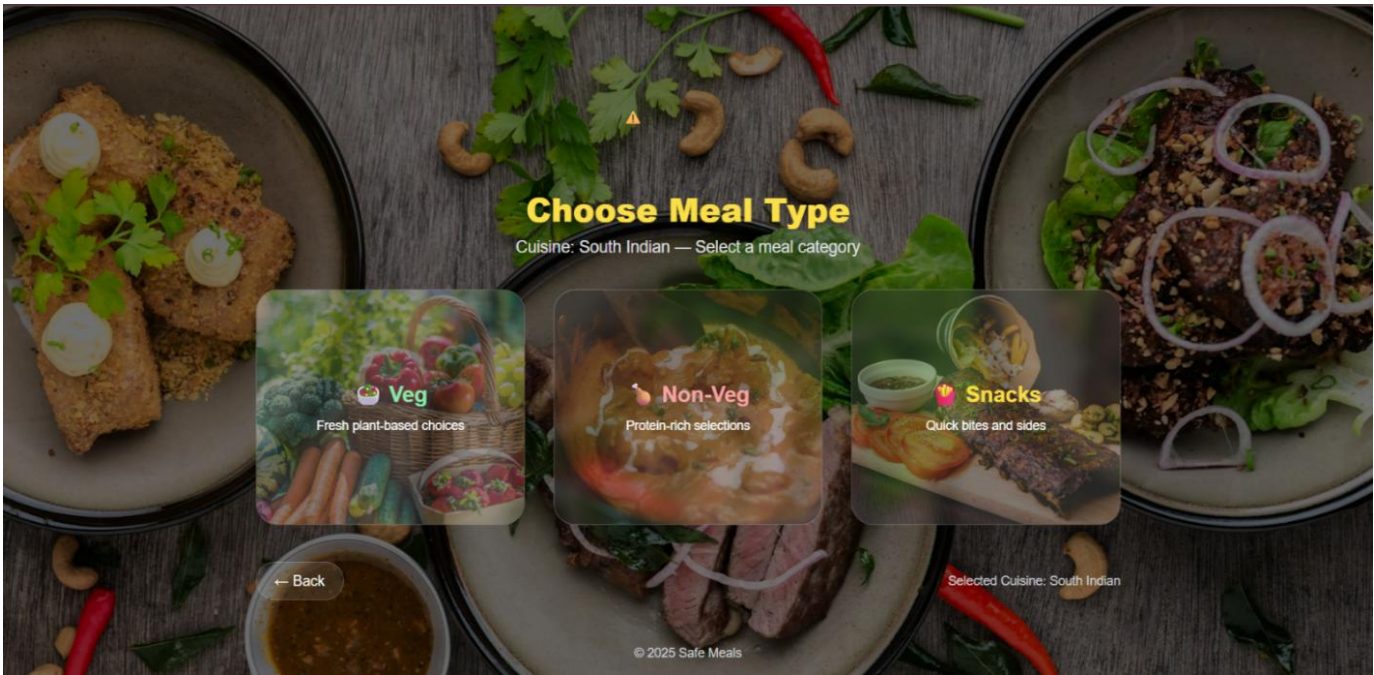
Login Page:



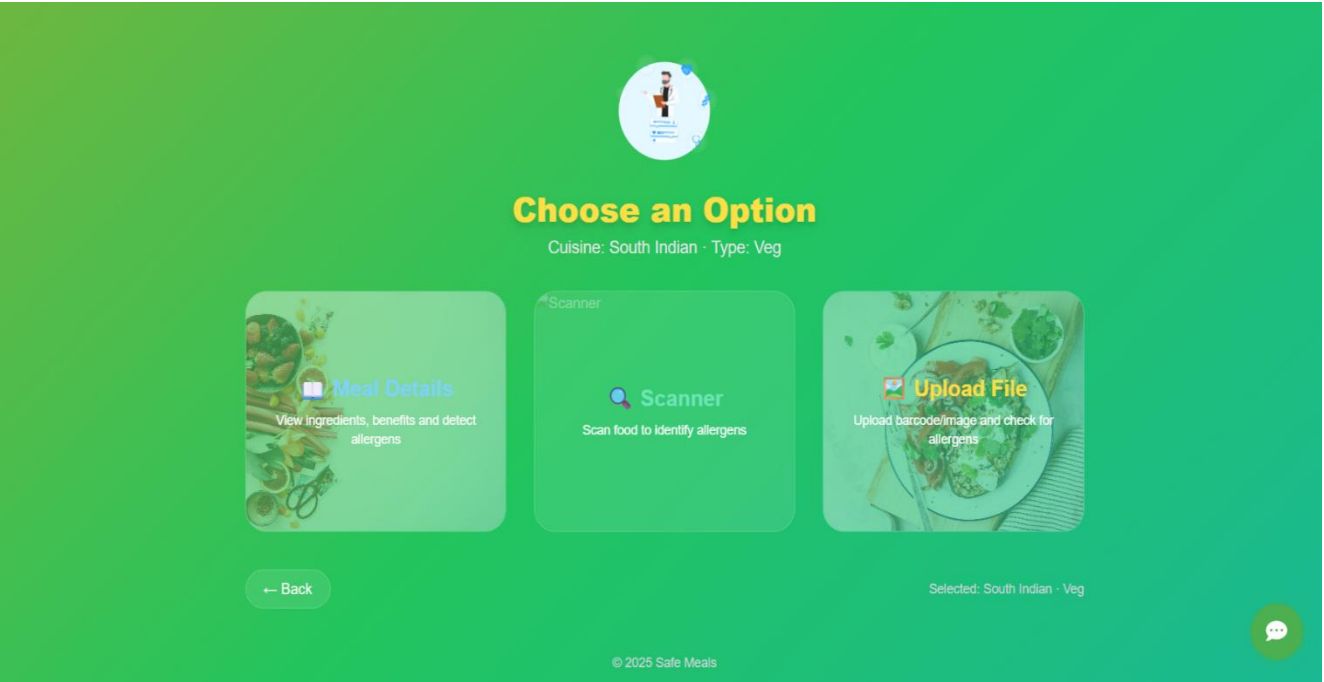
Cuisine Page:



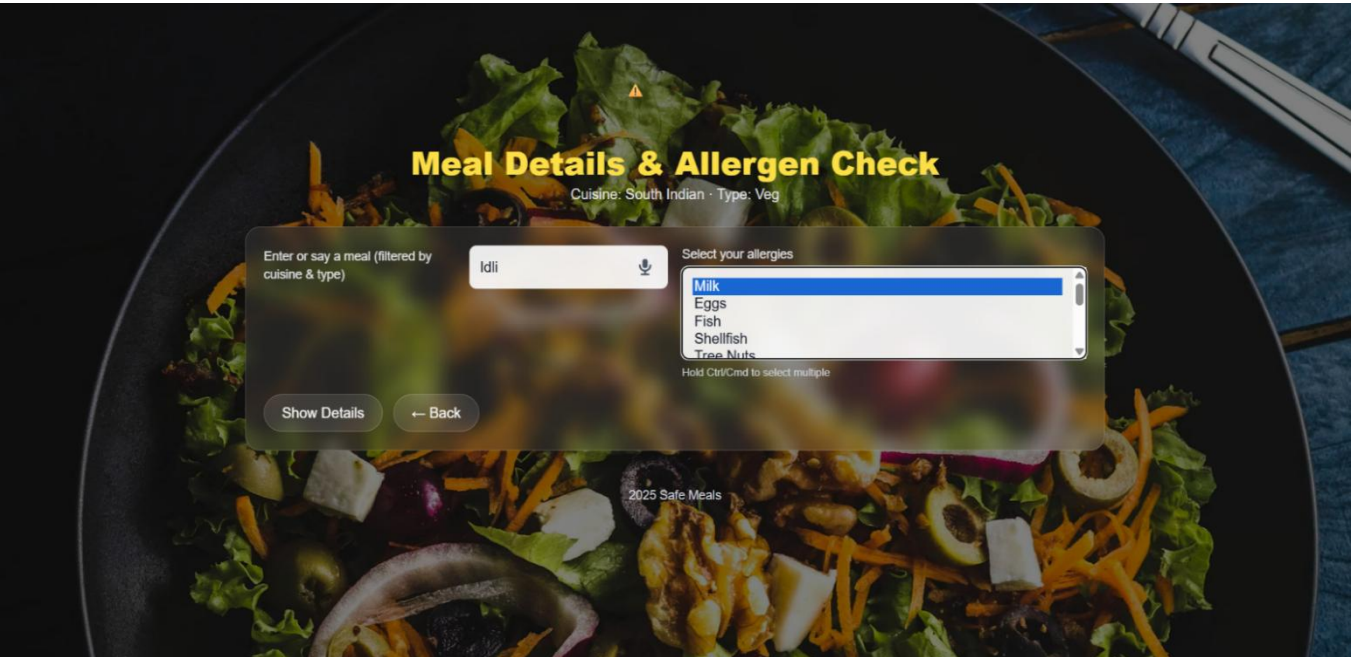
Meal Type Page:



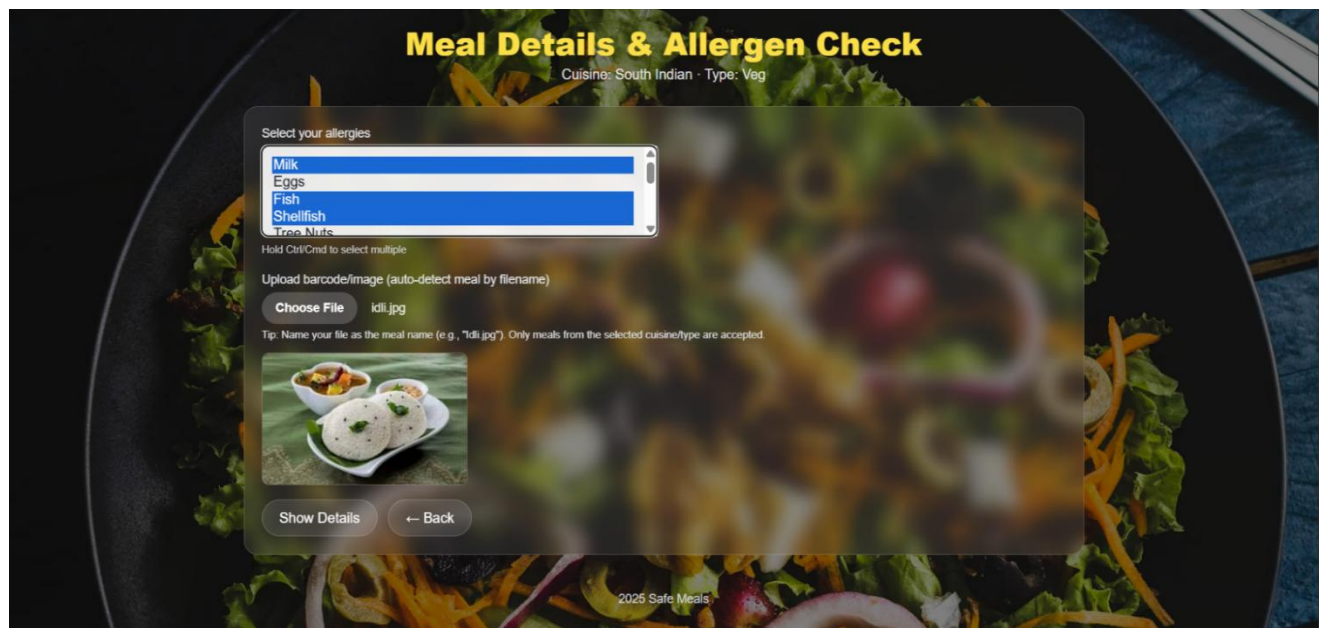
Option Page:



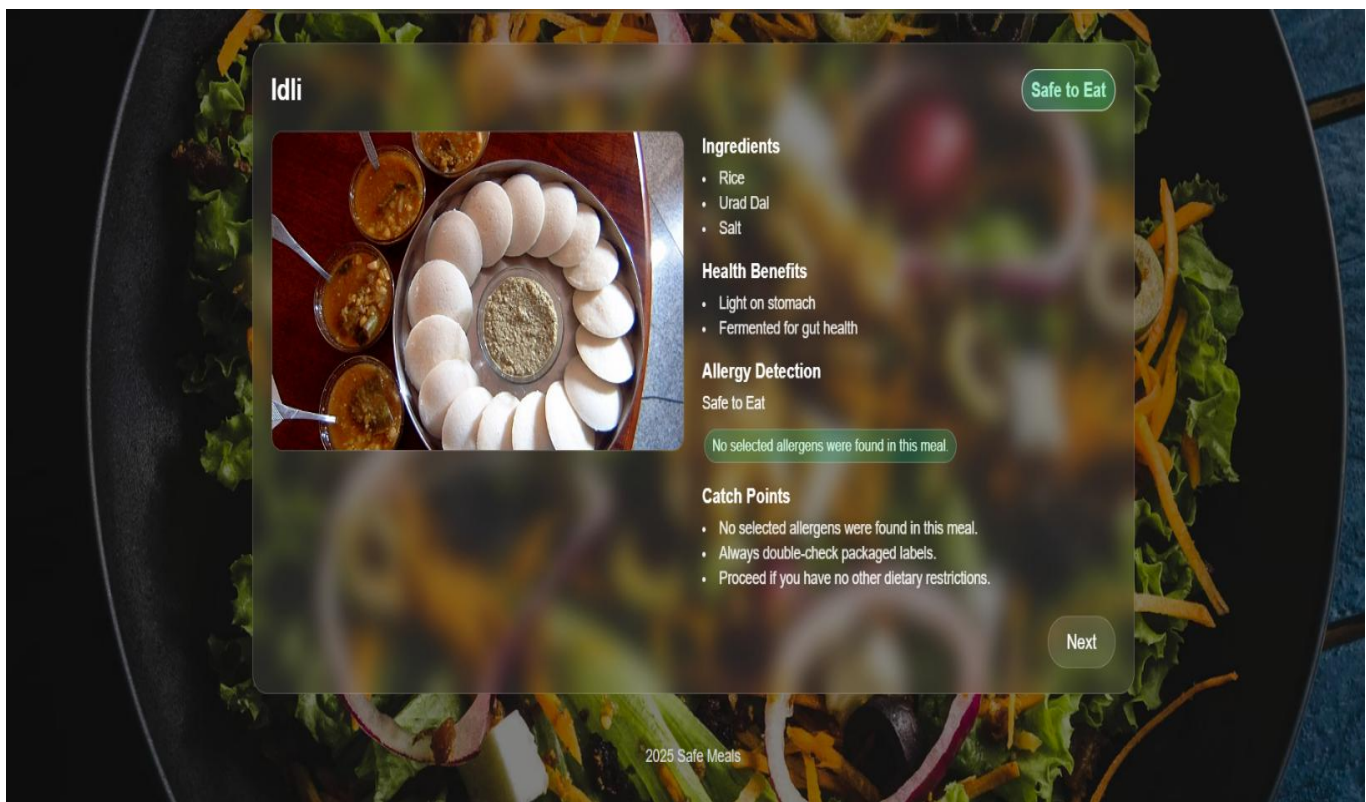
Meal Details Page:



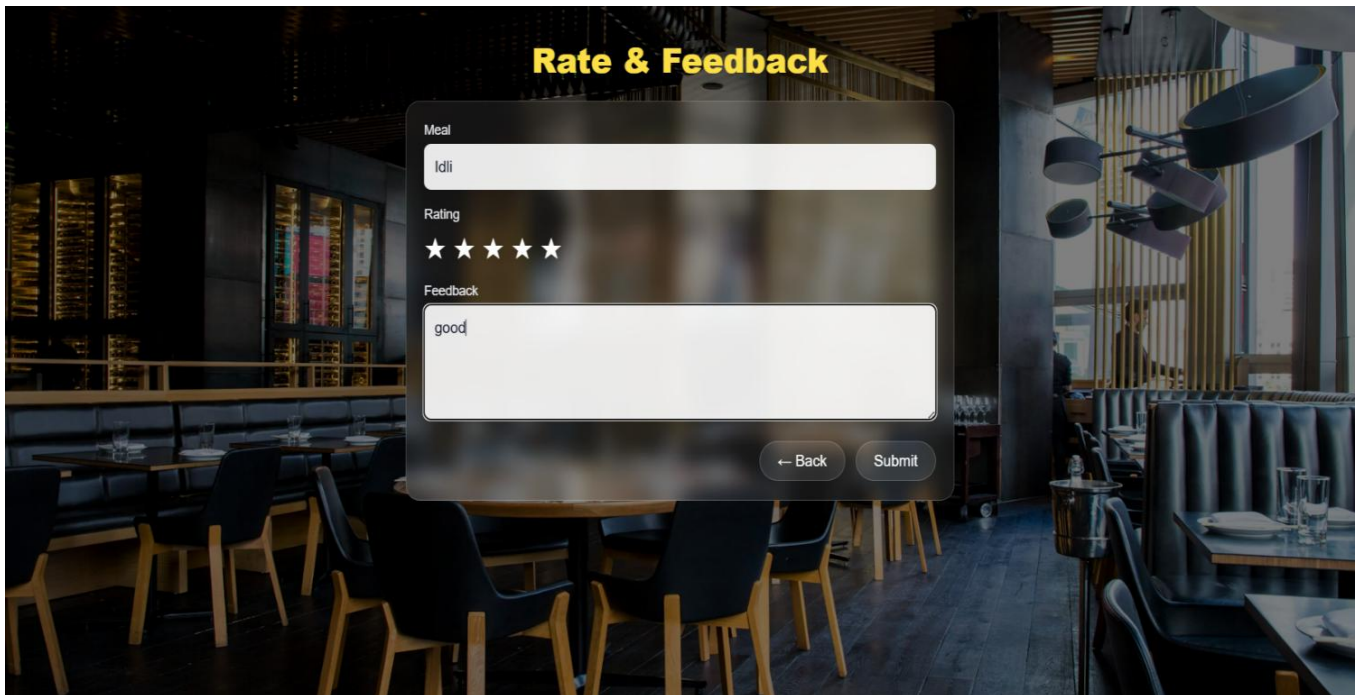
Upload Page:



Show Details Page:



Feedback Page:



Rate & Feedback

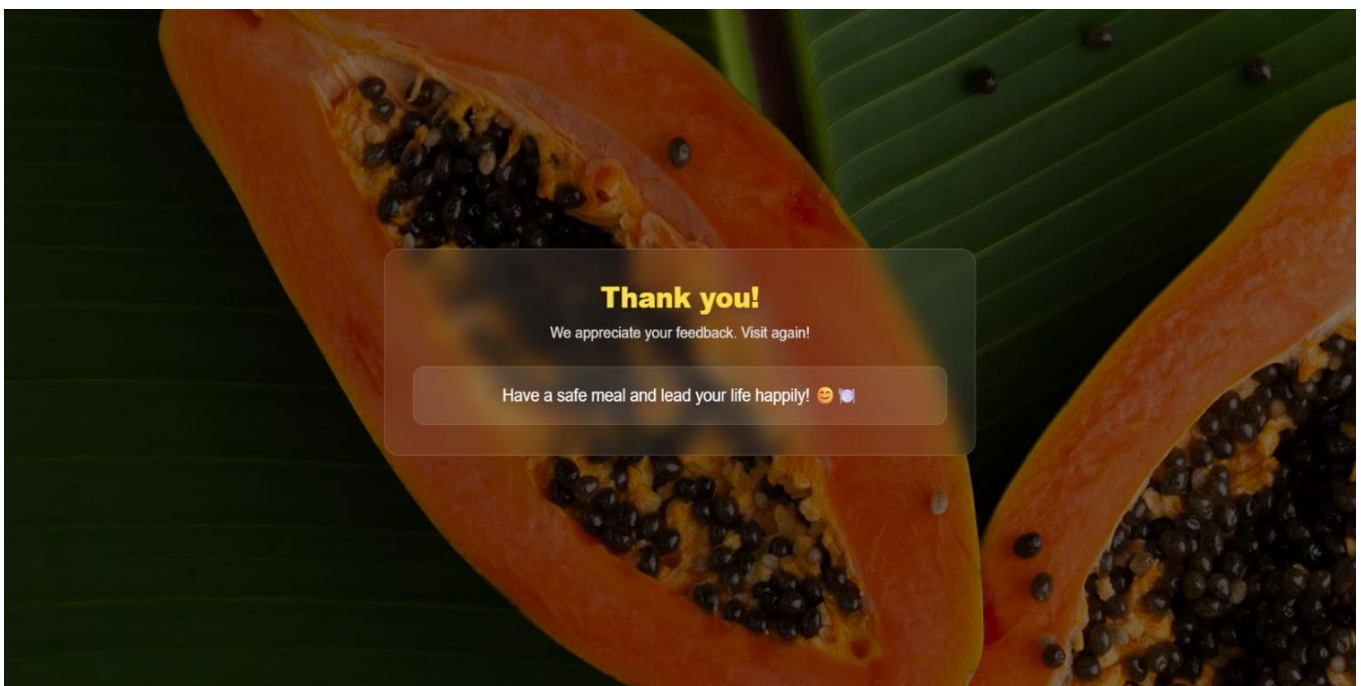
Meal
Idli

Rating
★ ★ ★ ★ ★

Feedback
good

← Back Submit

Thankyou Page:



Alert Email:

3 of 562

✓

Safe to Eat: Idli

Inbox x

Safe Meals

<thrishar54@gmail.com>

to me ▾

Wed, Nov 5, 11:44 AM (1 day ago)

☆

😊

↩

⋮

✓

Safe to Eat

Dear User,

Great news! **Idli** appears to be safe based on your allergy profile.

Stay safe and healthy!

Best regards,

The Safe Meals Team

⚠

Allergy Alert for Caesar Salad

Inbox

☆

S

Safe Me...

12:08 PM

😊

↩

⋮

to me ▾

⚠

Allergy Alert

Dear User,

We found potential allergens in **Caesar Salad** that match your profile:

• Fish

Please review the ingredients carefully before consuming.

Stay safe and healthy!

Best regards,

The Safe Meals Team

↩

Reply

➡

Forward

😊

99+

APPENDIX-III
INTERNSHIP CERTIFICATE

CERTIFICATE OF COMPLETION

INT:APP25-26/2887-3097

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Thrisha R, Reg. No: 310122104100**, a student of **Anand Institute Of Higher Technology**, pursuing **BE/CSE** has successfully completed an Internship Program at our organization in the domain of **UI/UX Design**.

The internship was undertaken from **11th August 2025 to 11th September 2025**.

During the course of the internship, the student exhibited commendable professional behaviour and technical proficiency, particularly in the project titled **"E Commerce Website"**.

We extend our best wishes to continue success in all future endeavors.

For Approtech R&D Solutions Pvt. Ltd.,


Authorized Signature

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